

Lesson 7

Chapter 3 (continued): continuity, utility maximization, and expenditure minimization

1 Continuity: no preference reversal

Continuity has the implication that there is no preference reversal: the ranking of two bundles cannot flip in the limit.

An equivalent definition

$\succsim$ is continuous if and only if, for every bundle y , the superior and inferior sets

$$\{x \in X : x \succsim y\} \quad \text{and} \quad \{x \in X : y \succsim x\}$$

are *closed*. Equivalently, their complements — the sets with strict preference, $\{x \in X : y \succ x\}$ and $\{x \in X : x \succ y\}$ — are *open*. In words: there is no dramatic change in the ordering if y changes a bit.

Why lexicographic preferences reverse: a worked example

Recall the sequence definition of continuity from Lesson 6: if $x^n \succsim y^n$ for every n , and $x^n \rightarrow x$, $y^n \rightarrow y$, then $x \succsim y$. A *preference reversal* is exactly a failure of this: the ranking holds all along the sequence but flips at the limit.

Take lexicographic preferences with good 1 compared first,

$$x \succ y \iff x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 > y_2.$$

Good 1 is infinitely more important than good 2: any extra amount of good 1, however small, beats any amount of good 2. Good 2 only matters to break an exact tie in good 1. Now fix

$$y = (1, 2) \quad (\text{so } y^n = y \text{ for all } n), \quad x^n = \left(1 + \frac{1}{n}, 1\right) \longrightarrow x = (1, 1).$$

1. *Along the sequence.* For every n , $x_1^n = 1 + \frac{1}{n} > 1 = y_1$, so good 1 decides and $x^n \succ y$. It does not matter that x^n has less of good 2 (1 against 2): the tiny surplus $\frac{1}{n}$ of good 1 outweighs it.
2. *At the limit.* $x = (1, 1)$ has exactly as much good 1 as y , so good 1 no longer decides. The tie is broken by good 2, and $1 < 2$, so $y \succ x$.
3. *The reversal.* $x^n \succ y$ for every n , yet $y \succ x$ for the limit. The surplus of good 1 shrinks smoothly to zero, but its *effect* on the ranking does not shrink: it stays decisive right up to the limit and then vanishes all at once, handing the decision to good 2. Continuity says exactly that this cannot happen, so lexicographic preferences are not continuous.

The same thing in terms of closed sets. Every x^n lies in the superior set of y , but the limit x does not. A set that does not contain the limits of its own sequences is not closed, so the superior set of y is not closed — in panel (b) of Figure 1, the missing piece is the dashed part of the vertical

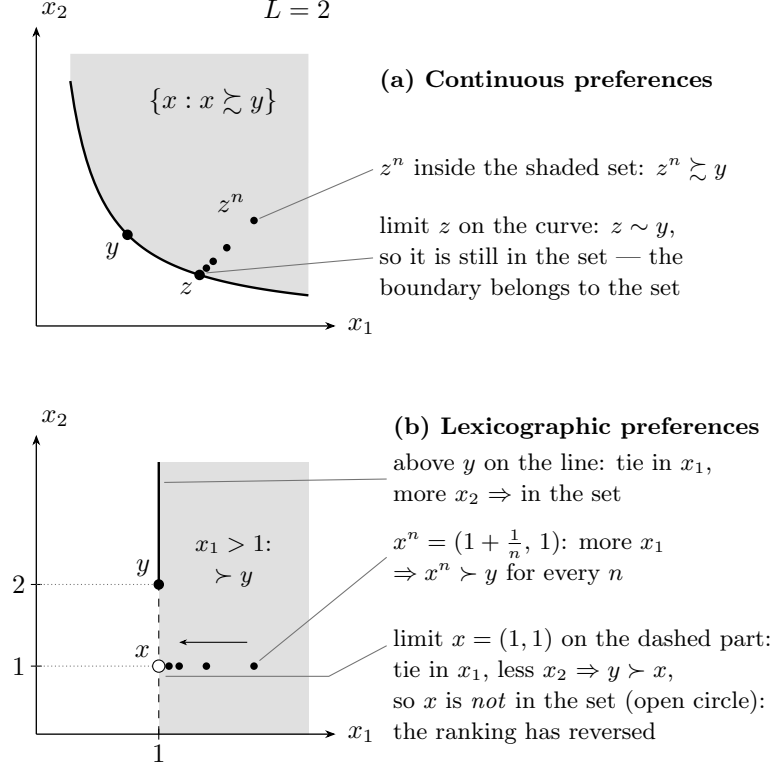


Figure 1: *Top*: with $L = 2$ goods and continuous preferences the superior set of y (shaded) contains its boundary, the indifference curve through y : it is closed, so a sequence z^n of bundles at least as good as y converges to a bundle z that is still at least as good as y . *Bottom*: the worked example below, with $y = (1, 2)$ and $x^n = (1 + \frac{1}{n}, 1) \rightarrow x = (1, 1)$ marked on the axes, for lexicographic preferences (good 1 compared first). The superior set of y is everything to the right of the vertical line through y , plus the part of that line *above* y ; the part below y (dashed) is not in it. The bundles x^n lie to the right, so $x^n \succ y$ for every n , but their limit x lies on the dashed part, so $y \succ x$: a preference reversal. The superior set is not closed.

line below y , where x sits. With continuous preferences (panel (a)) the boundary, the indifference curve, belongs to the set, and the limit of bundles that are at least as good as y is still at least as good as y .

Why the picture looks so odd. Under lexicographic preferences no two different bundles are indifferent, so there are no indifference curves at all: each “indifference set” is a single point. The same jump is behind the fact that no utility function can represent them: a utility u would have to give each vertical line $\{x_1 = a\}$ its own interval of values $(u(a, 0), u(a, 1))$, with the intervals of different lines not overlapping (everything on a line with more good 1 ranks higher). That would be uncountably many disjoint intervals in $\mathbb{R}$, each containing a different rational number — impossible, since there are only countably many rationals.

2 Utility representation

The main conclusion

If $\succsim$ is rational and continuous, there exists a *continuous* utility function $u(x)$ that represents $\succsim$ (Debreu's theorem). There is no utility function at all representing lexicographic preferences, which are rational but not continuous.

Differentiability

For calculus we need the extra assumption that $u(x)$ is differentiable. This is a genuine restriction: it rules out, for example, Leontief (perfect-complement) preferences,

$$u(x_1, x_2) = \min\{x_1, x_2\},$$

which are continuous but have a kink wherever $x_1 = x_2$.

Other connections between $\succsim$ and $u(x)$

$$\succsim \text{ is convex} \iff u(x) \text{ is quasiconcave,}$$

that is, $u(\alpha y + (1 - \alpha)z) \geq \min\{u(y), u(z)\}$ for all $\alpha \in [0, 1]$: every superior set $\{x : u(x) \geq c\}$ is convex. Likewise $\succsim$ is strictly convex if and only if u is strictly quasiconcave.

3 Utility maximization

Assume $u(x)$ exists and is continuous, and assume preferences are locally non-satiated. The *utility maximization problem* (UMP) is

$$\max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

- *Existence.* A solution exists, since $u(x)$ is continuous and, for $p \gg 0$ and $w \geq 0$, the budget set $B_{p,w}$ is non-empty and compact (Lesson 1), and a continuous function attains its maximum on a non-empty compact set (Weierstrass).
- The solution is the *Marshallian* (or *Walrasian*) demand $x(p, w)$.
- We may have many solutions, in which case we have a demand *correspondence*.

Some properties of $x(p, w)$

- *Walras' law*, $p \cdot x = w$ for every $x \in x(p, w)$: the budget binds, due to local non-satiation (Lesson 6).
- *Homogeneity of degree zero* in (p, w) : $x(\alpha p, \alpha w) = x(p, w)$ for $\alpha > 0$, because the feasible set does not change, $B_{\alpha p, \alpha w} = B_{p, w}$.

4 The indirect utility function

The indirect utility function is the maximum value function of the UMP,

$$v(p, w) = u(x(p, w)) :$$

the highest utility one can achieve at prices p and wealth w . Its properties:

- $v(p, w)$ is continuous in (p, w) (for $p \gg 0, w > 0$), as a consequence of the more general *theorem of the maximum*.
- $v(p, w)$ is homogeneous of degree zero in (p, w) , because the feasible set does not change.
- $v(p, w)$ is strictly increasing in w (by local non-satiation, more wealth always allows a better bundle) and non-increasing in p_i , for all $i = 1, \dots, L$ (a higher price only shrinks the budget set).
- $v(p, w)$ is *quasi-convex* in (p, w) : the inferior set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for every $\bar{v}$.

Why quasi-convex. Let $v(p, w) \leq \bar{v}$ and $v(p', w') \leq \bar{v}$, and put $(p'', w'') = \alpha(p, w) + (1 - \alpha)(p', w')$ with $\alpha \in [0, 1]$. Any x affordable at (p'', w'') satisfies $\alpha(p \cdot x) + (1 - \alpha)(p' \cdot x) \leq \alpha w + (1 - \alpha)w'$, so either $p \cdot x \leq w$ or $p' \cdot x \leq w'$: x was affordable in one of the two original situations, hence $u(x) \leq \bar{v}$. Taking the best such x , $v(p'', w'') \leq \bar{v}$.

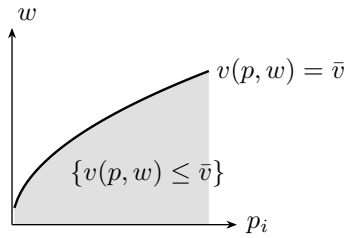


Figure 2: Quasi-convexity of v , drawn in the (p_i, w) plane with the other prices fixed. Along the curve utility stays at $\bar{v}$; below it (less wealth, or a higher price) utility is at most $\bar{v}$. That inferior set is convex, so the curve is concave.

The intuition

The slope of the curve $v(p, w) = \bar{v}$ is the extra wealth needed to keep utility at $\bar{v}$ when p_i rises: to first order, one unit of p_i costs x_i , the amount of good i being bought. The lower the price was to start with, the more of good i is bought (along the curve utility is held at $\bar{v}$, so this is the compensated response, which never goes the wrong way), so I need to be compensated more for a price change the lower the price was to start with. The curve is steep at low p_i and flattens out: it is concave.

5 More structure: uniqueness and differentiability

Impose more structure by assuming that $u(x)$ — which is *exogenous*, a primitive of the model — is strictly quasiconcave. Then the solution is *unique*, and $x(p, w)$ is a demand *function*. (Two different solutions would give the same utility, and by strict quasiconcavity their average — also affordable, since $B_{p,w}$ is convex — would give strictly more.)

Let us also assume that $x(p, w)$ and $v(p, w)$ are differentiable. Strictly speaking this is “a crime”: they are *endogenous*, derived from the model, so their properties should be proved rather than assumed. The good news is that we can now use a wonderful and elegant theorem, the *envelope theorem*.

6 The envelope theorem and Roy's identity

The envelope theorem tells us what happens to the maximum value, here the indirect utility, when a parameter changes: the derivative of the value function equals the partial derivative of the Lagrangian with respect to the parameter, evaluated at the optimum. The UMP

$$\max_x u(x) \quad \text{s.t.} \quad p \cdot x \leq w$$

has the Lagrangian

$$\mathcal{L} = u(x) - \lambda(p \cdot x - w).$$

The effect of wealth

$$\frac{\partial v(p, w)}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} \Big|_{\substack{x=x(p, w) \\ \lambda=\lambda^*}} = \lambda^* > 0.$$

The multiplier λ^* is the *marginal utility of wealth*; it is positive because v is strictly increasing in w .

The effect of a price

$$\frac{\partial v(p, w)}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} \Big|_{\substack{x=x(p, w) \\ \lambda=\lambda^*}} = -\lambda^* x_i(p, w) \leq 0.$$

Roy's identity

Combining the two (dividing the second by the first) gives

$$x_i(p, w) = - \frac{\partial v(p, w) / \partial p_i}{\partial v(p, w) / \partial w},$$

so we can infer demand from the indirect utility function.

7 The first-order conditions

Start from the Lagrangian and assume the solution is interior, $x \gg 0$. Then the Karush–Kuhn–Tucker (KKT) conditions are

$$\frac{\partial u(x)}{\partial x_i} - \lambda p_i = 0 \quad \Longleftrightarrow \quad \frac{\partial u(x)}{\partial x_i} = \lambda p_i, \quad i = 1, \dots, L,$$

together with $\lambda \geq 0$ and complementary slackness, $\lambda(w - p \cdot x) = 0$.

Is the constraint binding?

If $\partial u(x) / \partial x_i > 0$ for some good i — in particular if $\nabla u(x) \gg 0$ — then $\lambda^* = \frac{\partial u / \partial x_i}{p_i} > 0$, and by complementary slackness $p \cdot x = w$: the constraint is binding, as Walras' law says.

The tangency condition

Dividing the conditions for x_i and x_j , the multiplier cancels:

$$\frac{\partial u(x)/\partial x_i}{\partial u(x)/\partial x_j} = \frac{p_i}{p_j}.$$

The marginal rate of substitution equals the relative price. This is the *tangency condition*: at the optimum the indifference curve is tangent to the budget line.

8 Expenditure minimization

A thought experiment

The *expenditure minimization problem* (EMP) is

$$\min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u} :$$

the cheapest way of getting utility no less than $\bar{u}$ (for a level $\bar{u} > u(0)$). It is the *dual* of utility maximization: we switched the objective and the constraint. Instead of fixing the budget and reaching the highest indifference curve, we fix the indifference curve ($\bar{u}$) and find the lowest budget that reaches it.

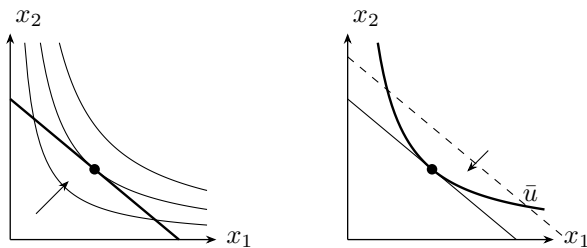


Figure 3: The two problems reach the same tangency from opposite sides. *Left, utility maximization*: the budget line is fixed and the consumer moves out to the highest indifference curve that touches it. *Right, expenditure minimization*: the indifference curve $\bar{u}$ is fixed and the budget line, keeping its slope $-p_1/p_2$, is shifted in until it just touches the curve.

When the two problems are matched — the EMP at $\bar{u} = v(p, w)$ — they pick the same bundle, and the minimum expenditure is exactly w .

9 The expenditure function

The expenditure function is the minimum value function of the EMP,

$$e(p, \bar{u}) = p \cdot h(p, \bar{u}),$$

where $h(p, \bar{u})$ denotes the solution of the EMP. Its properties (for $p \gg 0$):

- $e(p, \bar{u})$ is continuous in $(p, \bar{u})$.

- $e(p, \bar{u})$ is homogeneous of degree 1 in p : if all prices are scaled by α , relative prices are unchanged, so the solution is unchanged, $h(\alpha p, \bar{u}) = h(p, \bar{u})$. But the bundle now costs more:

$$e(\alpha p, \bar{u}) = \alpha e(p, \bar{u}), \quad \alpha > 0.$$

- $e(p, \bar{u})$ is increasing in $\bar{u}$ (a higher target costs more) and non-decreasing in p_i , for all $i = 1, \dots, L$.
- $e(p, \bar{u})$ is *concave* in p .

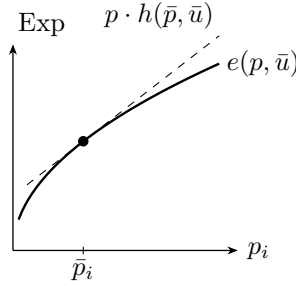


Figure 4: The expenditure function is concave in each price. The dashed line is the cost, at prices p , of the bundle $h(\bar{p}, \bar{u})$ that is optimal at $\bar{p}$: it is linear in p_i , touches e at $\bar{p}$, and lies above it everywhere else, because re-optimizing at the new prices can only lower the cost.

Proof of concavity

Let $p'' = \alpha p + (1 - \alpha)p'$ with $\alpha \in [0, 1]$, and let x'' denote the solution of the expenditure minimization problem at prices p'' . Then

$$\begin{aligned} e(p'', \bar{u}) &= p'' \cdot x'' = \alpha p \cdot x'' + (1 - \alpha)p' \cdot x'' \\ &\geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u}). \end{aligned}$$

The inequality holds because x'' reaches utility $\bar{u}$, so it is a feasible choice in the EMP at p and at p' ; the cheapest feasible bundle costs no more, $p \cdot x'' \geq e(p, \bar{u})$ and $p' \cdot x'' \geq e(p', \bar{u})$. This confirms concavity.

Transcription notes

- Page 1 of the original is a title page reading only “Lesson 7”; it is not reproduced.
- The direct utility function is written $V(x)$ throughout the original, and the indirect utility function $V(P, w)$ on some pages and $v(P, w)$ on others. Both are typeset in the course (MWG) notation: $u(x)$ for utility and $v(p, w)$ for indirect utility. Prices are typeset in lower case p ; the original writes P . The expenditure function, written $E(P, u)$, is typeset $e(p, \bar{u})$, with the utility target written $\bar{u}$ so that it is not confused with the function $u(\cdot)$.
- “equivilent” $\rightarrow$ *equivalent*; “Walra’s” $\rightarrow$ *Walras*; “Homongenity” $\rightarrow$ *homogeneity*; “continous” $\rightarrow$ *continuous*; “Lagrangian”, “Kuhn–Tucker”, “marginal” and “tangency” are spelled out where the original abbreviates or misspells them (“managin”, “tangent condition”); “KKT” is expanded as Karush–Kuhn–Tucker.
- **Corrected:** the set beside the first sketch on page 2 reads $\{x \in X \mid y \succ x\}$, which is the strict *inferior* set of y . The shading is above the indifference curve, and the sentence it illustrates is about the superior set being closed, so the label is typeset as the superior set $\{x : x \succsim y\}$.

- The second sketch on page 2 is the lexicographic order with good 1 compared first (the vertical line through y bounds the superior set), whereas the definition in Lesson 6 compares good 2 first. The labels “ $x^n \succ y^n$, $y \succ x$ ” are read with $y^n = y$, as drawn; the figure shows the sequence $x^n \rightarrow x$. **Added:** a worked numerical example, $y = (1, 2)$ and $x^n = (1 + \frac{1}{n}, 1) \rightarrow x = (1, 1)$, showing step by step how the ranking $x^n \succ y$ flips to $y \succ x$ at the limit, tied to both the sequence and the closed-set definitions of continuity. Figure 1 is stacked vertically and annotated: the coordinates of the example on the axes, a converging sequence $z^n \rightarrow z$ in the continuous panel, and callouts explaining each region and point.
- “Solution is manshallian on Walrasian demand” $\rightarrow$ *Marshallian or Walrasian demand*; “demand correspondence” $\rightarrow$ *correspondence*.
- “ $V(P, w)$ is homogeity degree zero” and “ $E(P, u)$ condition in (P, u) ” are read as *homogeneous of degree zero* and *continuous in $(p, \bar{u})$* .
- The sentence beside the quasi-convexity sketch, “I need to be compensated more for a price change the lower price was to start with”, is read as *the lower the price was to start with*.
- “Assume preferences are non-satiated” carries an insertion mark with “loc” above it; it is read as *locally non-satiated*.
- “I’m pose more structure” $\rightarrow$ *Impose more structure*. The remark “A crime” after assuming $x(p, w)$ and $v(p, w)$ differentiable is kept, with the reason it is a crime (they are endogenous) spelled out.
- **Corrected:** the tangency condition divides $\partial V / \partial x_i$ by $\partial V / \partial x_i$ in the original; the denominator is $\partial u / \partial x_j$, which is what gives p_j on the right.
- **Corrected:** the KKT condition is written $\partial V / \partial x_i = \partial P_i$ on the right of the “ $\iff$ ”; it is λp_i .
- **Corrected:** “ $E(P, w)$ is concave in P ” is written with w in place of the utility level; it is $e(p, \bar{u})$.
- On page 10 the monotonicity property reads “non-decreasing in u and in P_i ”. That is true, but in $\bar{u}$ the standard statement is stronger: with continuous, locally non-satiated preferences e is *strictly* increasing in $\bar{u}$; it is typeset as “increasing”.
- **Added:** the definition of quasiconcavity and the strict-convexity counterpart; the reason the UMP has a solution (Weierstrass); a proof that v is quasi-convex; the argument for uniqueness under strict quasiconcavity; the Lagrangian written out for the envelope theorem, with λ^* named the marginal utility of wealth; complementary slackness; the reason for the inequality in the concavity proof (feasibility of x''); “rational” in the representation theorem and the name Debreu; the countable-rationals argument for why lexicographic preferences have no utility representation (“Why the picture looks so odd”); the kink as the reason Leontief is not differentiable; the first-order reading of the slope of $v(p, w) = \bar{v}$ as x_i in “The intuition”; the restrictions $p \gg 0$, $w \geq 0$ or $w > 0$, and $\bar{u} > u(0)$; the remark that the UMP and EMP pick the same bundle when $\bar{u} = v(p, w)$; and, in the last figure, the dashed cost line of a fixed bundle, which shows why e is concave.
- The sketches were redrawn: the superior set for $L = 2$ and the lexicographic counter-example (page 2); the inferior set of v in the (p_i, w) plane (page 5); utility maximization against expenditure minimization (page 9); and $e(p, \bar{u})$ against p_i (page 10).
- Pages of the original, for tracing back: page 2 continuity, closed and open sets, and the two sketches; page 3 the representation theorem, lexicographic and Leontief preferences, convexity and quasiconcavity, and the UMP; page 4 existence, Marshallian demand, Walras’ law, homogeneity and $v(p, w)$; page 5 the properties of v and the quasi-convexity sketch; page 6 strict quasiconcavity, differentiability and the envelope theorem for w ; page 7 the envelope theorem for p_i , Roy’s identity and the KKT conditions; page 8 the binding constraint and the tangency condition; page 9 the EMP and its sketches; page 10 the expenditure function and its properties; page 11 the proof of concavity.